Fourier Transform

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$$\langle f,g \rangle = \int_a^b f(x)\, g(x)\, dx$$

orthogonality : $\langle f,g \rangle = 0$

normal : $\|f\| = \sqrt{\langle f,f \rangle} = 1$

orthonormality : $\|f\| = \|g\| = 1$ and $\langle f,g \rangle = 0$

Fourier's Theorem: suppose f is periodic w/ period $2L$, $f, f' \in PC[-L,L]$

$$f(x) = \frac{a_0}{2} + \sum_{m=1}^{\infty} \left(a_m \cos\frac{m\pi x}{L} + b_m \sin\frac{m\pi x}{L}\right)$$

$$a_m = \left\langle f, \frac{1}{L}\cos\frac{m\pi x}{L} \right\rangle = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{m\pi x}{L}\, dx$$

$$b_m = \left\langle f, \frac{1}{L}\sin\frac{m\pi x}{L} \right\rangle = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{m\pi x}{L}\, dx$$

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\, dx$$

PDE's

BVP:
$$\begin{cases} \frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2} \\ u(x,0) = f(x) \\ u(0,t) = u(L,t) = 0 \end{cases} \qquad \begin{matrix} 0<x<1 \\ t>0 \end{matrix}$$

assume soln separable

$u(x,t) = X(x)\,T(t)$

$u_t = XT'$

$u_{xx} = X''T$

$XT' = \alpha^2 X'' T$

$\frac{T'}{\alpha^2 T} = \frac{X''}{X} = -\lambda$

$X'' = -\lambda X$

↳ $X(t) = A\cos(\sqrt{\lambda}x) + B\sin(\sqrt{\lambda}\,x)$

$X(0) = 0 \Rightarrow A = 0$

$X(L) = 0 \Rightarrow B\sin\sqrt{\lambda}L = 0$

$\sqrt{\lambda} = \frac{n\pi}{L}, \quad n \in \mathbb{R}$

$\underline{\lambda_n = \frac{n^2\pi^2}{L^2}}$

$X_n(x) = \sin\frac{n\pi x}{L}$

$T' = -\lambda\alpha^2 T \Rightarrow T' = -\frac{n^2\pi^2}{L^2}\alpha^2 T$

$T_n(t) = \exp\left(-\frac{n^2\pi^2\alpha^2}{L^2}t\right)$

$u_n(x,t) = X_n(\lambda)\,T_n(t)$

general soln : $u(x,t) = \sum_{n=0}^{\infty} c_n \sin\frac{n\pi x}{L}\exp\left(-\frac{n^2\pi^2\alpha^2}{L^2}t\right)$

Fourier Transform

$$F[f](w) = \hat{f}(w) = \int_{-\infty}^{\infty} f(x)\, e^{-iwx}\, dx$$

① Linearity: $\widehat{\alpha f + g} = \alpha\hat{f} + \hat{g}$

② Derivatives: $F\left[\frac{df}{dx}\right] = iw\hat{f}(w)$

③ Convolutions: $F[f+g](w) = \hat{f}\cdot\hat{g}$

④ Translations: $\widehat{f(x-\alpha)} = e^{-iw\alpha}\hat{f}(w)$

Solving PDE's

$$\begin{cases} \frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2} \\ u(x,0) = f(x) \end{cases} \qquad x \in \mathbb{R},\ t>0$$

Take Fourier transform in x:

$$\hat{u}(w,t) = \int_{-\infty}^{\infty} u(x,t)\, e^{-iwx}\, dx$$

$$\widehat{\frac{\partial u}{\partial t}} = \int_{-\infty}^{\infty} \frac{\partial u}{\partial t} e^{-iwx}\, dx = \frac{\partial}{\partial t}\int_{-\infty}^{\infty} u\, e^{-iwx}\, dx = \frac{\partial}{\partial t}\hat{u}$$

$$\widehat{\frac{\partial^2 u}{\partial t^2}} = (iw)^2\hat{u} = -w^2\hat{u}$$

$\frac{\partial}{\partial t}\hat{u} = -\alpha w^2 \hat{u}$

↳ $\hat{u}(w,t) = \hat{u}(w,0)\, e^{-\alpha w^2 t} = \hat{f}(w)\, e^{-\alpha w^2 t}$

$u(w,t) = F^{-1}[\hat{u}(w,t)](\lambda)$